EXERCISE

1. Encircle the correct choice

(i) The slight negative charge at one end of one water molecule is attracted to the slight positive charge of another water molecule. What is this attraction called?
(a) Covalent bond (b) Hydrogen bond
(c) Ionic bond (d) Hydrophilic bond

(ii) Tendency of water to coalesce oil drop into large droplet called
(a) Hydrophilic force (b) Hydrophobic exclusion
(c) Hydrophilic exclusion (d) Hydrogen bonding

(iii) The covalent bond between two monosaccharides is called
(a) Peptide bond (b) Ester bond
(c) Phosphodiester bond (d) Glycosidic bond

(iv) Most abundant carbohydrate found in nature?
(a) Glucose (b) Maltose
(c) Cellulose (d) Glycogen

(v) Most important organic compound of the cell which carry out virtually all of the cell's activities.
(a) Protein (b) Carbohydrates
(c) Nucleic acid (d) Lipids

(vi) All amino acids have same formula except
(a) Alpha carbon (b) Hydroxyl group
(c) Radical group (d) Amino group

(vii) A trihydroxy alcohol, made of three carbon atoms called
(a) Glucose (b) Glycerol
(c) Maltose (d) Ribose

(viii) Large and important class of lipid made up of isoprenoid unit called
(a) Phospholipid (b) Terpenoids
(c) Waxes (d) Acylglycerol

(ix) How many molecules of water are needed to completely hydrolyze a polymer that is 21 monomers long?
(a) 10 (b) 20
(c) 21 (d) 2

(x) Which of the following is true of both starch and cellulose?
(a) They are both polymers of glucose
(b) They are geometric isomers of each other.
(c) They can both be digested by humans
(d) They are both used for energy storage in plants.

- Oxido-reductase, Transferase, Hydrolase, Lysase, Isomerase, Ligase are the type of enzyme on the basis of reactions.
- Protease, Lipase, Glycosidase, Nuclease are the types of enzyme on the basis of substrate on which enzyme act.

EXERCISE

1. Encircle the correct choice

(i) Which of the following statements is (are) true about enzyme catalyzed reactions?

(a) The reaction is faster than the same reaction in the absence of the enzyme.
(b) The free energy change of the reaction is opposite from the reaction in the absence of the enzyme.
(c) The reaction always goes in the direction toward chemical equilibrium.
(d) Both a and b.

(ii) The active site of an enzyme is the region that

(a) Binds allosteric regulators of the enzyme
(b) Is involved in the catalytic reaction of the enzyme
(c) Binds the products of the catalytic reaction
(d) Is inhibited by the presence of a coenzyme or a cofactor.

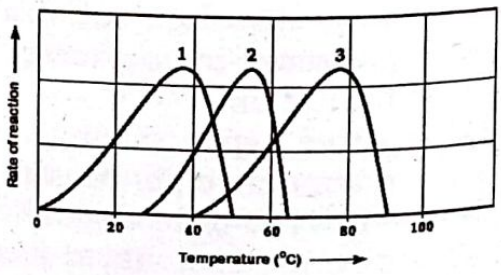
(iii) According to the induced fit hypothesis of enzyme catalysis, which of the following is correct?

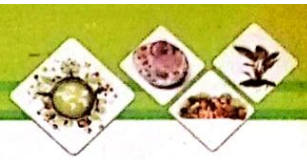
(a) The binding of the substrate depends on the shape of the active site.
(b) Some enzymes change their structure when activators bind to the enzyme.
(c) A competitive inhibitor can outcomepete the substrate for the active site.
(d) The binding of the substrate changes the shape of the enzyme's active site.

(iv) Which curve represents the behavior of an enzyme taken from a bacterium that lives in hot-springs.

I. Curve 1 II. Curve 2 III. Curve 3

(a) I only
(b) II only
(c) III only
(d) I and II
(e) II and III





(v) Increasing the substrate concentration in an enzymatic reaction could overcome which of the following?
(a) Denaturization of the enzyme
(b) Allosteric inhibition
(c) Competitive inhibition
(d) Saturation of the enzyme activity

(vi) Competitive inhibitors block the entry of substrate into the active site of an enzyme. On which of the following properties of an active site does this primarily depend?
(a) The ability of an enzyme to form a template for holding and joining molecules
(b) The enzyme's ability to stretch reactants and move them toward a transition state
(c) The enzyme providing an appropriate microenvironment conducive to a reaction's occurrence
(d) The enzyme forming covalent bonds with the reactants

(vii) A series of enzymes catalyze the reaction $X \rightarrow Y \rightarrow Z \rightarrow A$. Product A binds to the enzyme that converts X to Y at a position remote from its active site. This binding decreases the activity of the enzyme.What is substance X?
(a) A coenzyme
(b) An allosteric inhibitor
(c) A substrate
(d) An intermediate

(viii) If an enzyme is added to a solution where its substrate and product are in equilibrium, what would occur?
(a) Additional product would be formed.
(b) Additional substrate would be formed.
(c) The free energy of the system would change.
(d) Nothing; the reaction would stay at equilibrium.

(ix) Which one of the following is the mechanism of action of enzymes?
(a) They act upon substrate molecules to release new substrate molecules.
(b) They actually increase the amount of energy of activation.
(c) Enzymes dramatically decrease the amount of energy of activation.
(d) Enzymes break product molecules to release new product molecules.



(x) How does a noncompetitive inhibitor decrease the rate of an enzyme reaction?

(a) By binding at the active site of the enzyme
(b) By changing the shape of a reactant
(c) By changing the free energy change of the reaction
(d) By acting as a coenzyme for the reaction

2. Write short answers of the following questions:

1. Why enzymes are specific in nature?
2. Why enzyme activity is directly proportional to enzyme concentration?
3. Why enzymes are called temperature sensitive?
4. How enzyme reduces the energy of activation?
5. How enzymes activity effect on substrate concentration?
6. What do you mean by prosthetic group?
7. Differentiate between the activator and inhibitors.

3. Write detailed answers of the following questions:

1. What is enzyme? explain characteristics of enzyme.
2. Describe the effect if temperature on the rate of enzyme action.
3. Describe mechanism of enzyme action.
4. Describe the classification of enzymes.
5. Describe enzymatic inhibition, its types and its significance.
6. Describe structure of enzyme.
7. Explain the effect of substrate and concentration on the rate of enzyme action.

- Glyoxysome contain enzymes that initiate the conversion of fatty acid into sugar.
- Mitochondria have a semi-autonomous existence in the cell.
- The most common type of plastid, containing chlorophyll which gives green colour to plants is the site of photosynthesis.
- A network of different protein fibers which provide three dimensional shapes to the cell called cytoskeleton.
- Centrioles are short, barrel shaped structure of microtubules.
- The main function of central vacuole is to maintain turgor pressure inside plant cells.

EXERCISE

1. Encircle the correct choice

(i) A primary objective of cell fractionation is to
(a) View the structure of cell membranes.
(b) Identify the enzymes outside the organelles.
(c) Determine the size of various organelles.
(d) Separate the major organelles so that their particular functions can be determined.

(ii) The volume enclosed by the plasma membrane of plant cells is often much larger than the corresponding volume in animal cells. The most reasonable explanation for this observation is that
(a) Plant cells are capable of having a much higher surface to volume ration than animal cells.
(b) Plant cells have a much more highly convoluted plasma membrane than animal cells.
(c) Plant cells contain a large vacuole that reduces the volume of the cytoplasm.
(d) Animal cells are more spherical, while plant cells are elongated.

(iii) Large numbers of ribosomes are present in cells that specialize in producing which of the following molecules?
(a) Lipids (b) Starch
(c) Proteins (d) Steroids

(iv) In animal cells, hydrolytic enzymes are packed to prevent general destruction of cellular components. Which of the following organelles functions in the compartmentalization?
(a) Chloroplast (b) Lysosome
(c) Peroxisome (d) Glyoxysome



(v) Tay-Sachs disease is a human genetic abnormality in cells accumulating and becoming clogged with very large and complex lipids. Which cellular organelle must be involved in this condition?
(a) Endoplasmic reticulum (b) Golgi complex
(c) Lysosome (d) Mitochondria

(vi) Which is one of the main energy transformers of cells?
(a) Endoplasmic reticulum (b) Golgi complex
(c) Lysosome (d) Mitochondria

(vii) Organelles other than the nucleus that contain DNA
I. Ribosomes II. Chloroplast III. Mitochondria
(a) I only (b) II only
(c) II and III (d) I and II

(viii) Which structure is common to plant and animal cells?
(a) Chloroplast (b) Cell wall
(c) Central vacuole (d) Mitochondria

(ix) Cell organelle mainly concerned with the detoxification of alcohol.
(a) Chloroplast (b) Peroxisome
(c) Central vacuole (d) Mitochondria

(x) Clarity of image is generally known as
(a) Magnification (b) Contrast
(c) Resolution (d) Sedimentation

2. Write short answers of the following questions:

1. Why lysosome is called suicidal sacs?
2. Why plasma membrane is differentially permeable in nature?
3. Why plant cell wall is rigid?
4. Why chloroplast is called energy converting cell organelle?
5. How prokaryotic ribosome is different from eukaryotic ribosome.
6. How mitochondria is similar to bacteria?
7. Why mitochondria is called power house of cell?
8. Differentiate between peroxisome and glyoxysome.

3. Write detailed answers of the following questions:

1. Describe structure and functions of rough and smooth endoplasmic reticulum.
2. Explain the chemical composition and functions of plasma membrane in regulating cell's interactions with environment.
3. Explain the structure and functions of lysosomes.
4. Describe structure of mitochondria with suitable diagram.
5. Explain the structure and composition of cell wall.
6. Describe the structure and functions of Golgi complex.
7. Describe the types, structure, composition and functions of cytoskeleton.



EXERCISE

1. Encircle the correct choice

(i) Chlorophyll-a is almost identical to chlorophyll-b but slight structural difference between them is enough to give
- (a) Similar energy during the light reaction
- (b) Different absorptive spectrum
- (c) Different product during the Calvin cycle
- (d) All of these

(ii) Chlorophyll is organized along with other molecules into photosystem, which has light gathering
- (a) Reaction Centre
- (b) Carotenoid compound
- (c) Antenna complex
- (d) Cytochrome

(iii) Select the correct statement
- (a) PSI and ATP synthase complexes are located in the appressed part of thylakoid.
- (b) PSI and NADP reductase are located in the appressed part of thylakoid membrane
- (c) Appressed part Contain NADP reductase and ATP synthase
- (d) Non appressed (non-stacked) Having PSI

(iv) The linear flow of electrons from water to $NADP^+$ coupled to ATP synthesis is
- (a) Cyclic photophosphorylation
- (b) Non-Cyclic photophosphorylation
- (c) Chemiosmotic phosphorylation
- (d) Oxidative phosphorylation

(v) The intermediate carbon fixing compound in the member of grass family to pass CO_2 to calvin cycle is
- (a) Citric acid
- (b) Oxaloacetic acid
- (c) Pyruvic acid
- (d) Crassulacean acid

(vi) Oxidative decarboxylation of isocitrate form?
- (a) α-ketoglutarate
- (b) Succinate
- (c) Cis-aconitate
- (d) Fumarate

(vii) How many ATP molecules are forms during substrate level phosphorylation in kreb's cycle when one glucose is consumed?
- (a) One
- (b) Two
- (c) Three
- (d) Four



(viii) Enzyme involves during carboxylation
- (a) Rubisco oxygenase
- (b) Rubisco carboxylase
- (c) Rubisco dehydrogenase
- (d) No need of enzyme during carboxylation

(ix) The oxygen consumed during cellular respiration is involved directly in which process or events?
- (a) Glycolysis
- (b) Accepting electrons at the end of the electron transport chain
- (c) The citric acid cycle
- (d) The oxidation of pyruvate to acetyl CoA

(x) How many carbon atoms are fed into the citric acid cycle as a result of the oxidation of one molecule of pyruvate?
- (a) 2 (b) 4
- (c) 6 (d) 8

2. Write short answers of the following questions:

1. Why antenna complex contains other pigments with chlorophyll?
2. Why photosynthesis is called redox process?
3. How cyclic photophosphorylation helpful in photosynthesis?
4. Why ATP is called common energy currency of living system?
5. Why Calvin cycle is also called C3 cycle?
6. Why CAM plant close stomata in day time?
7. Why oxidation of pyruvate provide more energy than lactic acid fermentation?

3. Write detailed answers of the following questions:

1. Explain in detail light independent phase of photosynthesis.
2. What is cellular respiration? Explain types of respiration in detail.
3. Explain event takes place in breaking of glucose in cytosol.
4. Discuss cyclic and non-cyclic photophosphorylation during light reaction.
5. Describes tricarboxylic acid cycle in detail.
6. Explain alternative mechanism of CO_2 fixation in plant
7. Explain chemiosmosis and oxidative phosphorylation.



EXERCISE

1. Encircle the correct choice.

(i) Viruses evolve by using the
(a) Cellular organelles (b) Cellular enzymes
(c) Cellular energy (d) All of them

(ii) The pathogenicity of a virus depends upon
(a) the immunity of the host body
(b) the effective penetration of its genome
(c) the overall environment inside the body
(d) the overall environment outside the body

(iii) Bacteriophages escape from host cell by the activity of
(a) Lysozyme (b) Ribozyme
(c) Peroxisomes (d) Glyoxisomes

(iv) The smaller proteins are cut down and forms a new virus structure by the process called
(a) Integration (b) Transcription
(c) Budding (d) Assembly

(v) Virus that severely damage motor neurons and causes paralysis called
(a) HIV (b) Dengue
(c) Polio (d) Herpes

(vi) Proteins that cause pathogenicity in humans and animals called
(a) Prions (b) Viroids
(c) Antigen (d) Antibodies

(vii) Smaller than viruses having single stranded RNA with some double stranded regions are called
(a) Prions (b) Viroids
(c) Minus strand viruses (d) Double stranded DNA viruses

(viii) Dengue fever, encephalitis and yellow fever are caused by which group of viruses?
(a) Arbo-viruses (b) Retro-viruses
(c) Rhabdo-viruses (d) Rhino-viruses

(ix) Aedes mosquito is the vector of
(a) Dengue virus (b) Ebola virus
(c) Hepatitis virus (d) Measles virus



EXERCISE

1. Encircle the correct choice

(i) Which of the following is true about the Kingdom Protoctista
- (a) All organisms are prokaryotes
- (b) All organisms are eukaryotes
- (c) All organisms are autotrophic
- (d) All organisms are heterotrophic

(ii) All members of green algae have the pigment combination
- (a) Chlorophylls a and b
- (b) Chlorophylls a and c
- (c) Chlorophyll a and Phycobilins
- (d) Chlorophylls a , b and c

(iii) The major grouping of protozoa is based upon their
- (a) Feeding habits
- (b) Mode of reproduction
- (c) Mode of Nutrition
- (d) Mode of locomotion

(iv) Protoctista having isomorphic alternation of generation
- (a) Chlorella
- (b) Ulva
- (c) Euglena
- (d) Phytophthora

(v) The amoeboid stage of slime mold is
- (a) Plasmodium
- (b) Fruiting bodies
- (c) Euglena
- (d) Merozoites

(vi) Trypanosomes and Leishmanias are disease causing
- (a) Viruses
- (b) Bacteria
- (c) Protoctists
- (d) Fungi

(vii) Ulva is commonly called
- (a) Bacteria
- (b) Virus
- (c) Mushroom
- (d) Sea lettuce

(viii) The study of fungi is called
- (a) Zoology
- (b) Microbiology
- (c) Botany
- (d) Mycology

(ix) The most important cellulose degraders in ecosystem are
- (a) Ascomycota
- (b) Zygomycota
- (c) Basidiomycota
- (d) Deutromycota



EXERCISE

1. Encircle the correct choice

(i) The rapid diversification of plants occurred in the period:
(a) Silurian (b) Devonian
(c) Carboniferous (d) Paleozoic

(ii) Spores of land plants tough, flexible and resistant to biodegradation by:
(a) Lignin (b) Chitin
(c) Waxes (d) Sporopollenin

(iii) Hepaticae, Anthocerotae and Moss are the classes of:
(a) Angiosperms (b) Gymnosperms
(c) Tracheophytes (d) Bryophytes

(iv) The most successful kind of reproduction is:
(a) Heterogamy (b) Oogamy
(c) Anisogamy (d) Isogamy

(v) The dominant generation of living vascular plants existing as free-living plants is:
(a) Sporophyte (b) Gametophyte
(c) Tracheophyte (d) Pteridophyte

(vi) The most primitive living vascular plants is:
(a) Lycophyta (b) Sphenophyta
(c) Psilophyta (d) Filicinophyta

(vii) The gametophyte of Fern is called:
(a) Archegonium (b) Antheridium
(c) Cones (d) Prothallus

(viii) Double fertilization is the characteristic feature of:
(a) Bryophytes (b) Gymnosperms
(c) Angiosperms (d) Ferns

(ix) Resins, turpentine and pine oil are obtained from:
(a) Rose (b) Onion
(c) Pins (d) Banana

(x) In banana tree, flowers are covered over by one or many large brackets called:
(a) Spathes (b) Spadix
(c) Capitulum (d) Palea



EXERCISE

1. Encircle the correct choice.

(i) Choose the incorrect pair?
- (a) Sponges → Spicules.
- (b) Cnidaria → Nematocycts.
- (c) Segmented worm → Psuedopodia.
- (d) Arthropoda → jointed legs.

(ii) Water movement through sponge?
- (a) Ostia → Spongocoel → Osculum.
- (b) Spongocoel → Osculum → Ostia.
- (c) Ostia→ Osculum → Spongocoel.
- (d) Osculum → Spongocoel → Ostia.

(iii) Choose the term which encompasses all other in the list
- (a) Coelomate
- (b) Protostome
- (c) Bilateria
- (d) Triploblastic

(iv) Dry scaly skin, ectothermic, two pairs of legs, lungs, Internal Fertilization, eggs with amniotic shell, and a three chambered heart are all properties of which class?
- (a) Amphibia
- (b) Reptilia
- (c) Aves
- (d) Mammals

(v) Pick the odd one out?
- (a) Star fish.
- (b) Brittle star.
- (c) Lamprey.
- (d) Sea urchin.

(vi) Which of the following terms or structures is properly associated only with animals?
- (a) Genes.
- (b) Cell wall.
- (c) Autotrophy.
- (d) Sexual reproduction.

(vii) An adult animal that possesses bilateral symmetry is most certainly also
- (a) Triploblastic.
- (b) Deuterostomes.
- (c) Coelomate.
- (d) Protostomes.

(viii) Protostome characteristics generally include which of the following?
- (a) A Mouth that develops secondly and far away from the blastopore
- (b) Radial body
- (c) Radial cleavage.
- (d) Determinate cleavage

(ix) Sponges structural materials (spicules) are manufactured by the
- (a) Pinacocytes.
- (b) Choanocytes.
- (c) Amoebocytes.
- (d) Porocytes.

(x) Which of the following can be used to distinguish a nematode from annelids?

I. Type of body cavity
II. Number of muscle layers in the body wall
III. Presence of segmentation
IV. Number of embryonic tissue layers

(a) I only (b) II only
(c) I and II (d) I, II and III

2. Write short answers of the following questions:

1. How animals are classified?
2. Why sponges placed in animal kingdom, when it lacks tissue organization?
3. Why flat worms are named so?
4. Differentiate between invertebrate chordates and vertebrates
5. Differentiate between cartilaginous fishes and bony fishes
6. Why protostomes are named so?
7. Why diploblastic animals are different from the triploblastic?
8. Why echinoderms are placed in bilateria when adults are radially symmetrical?

3. Write detailed answers of the following questions:

1. Discuss the basic factors which helps in classification of animals
2. Write down evolutionary adaptations of phylum annelida.
3. Write down the general characters of phylum arthropoda
4. Describe the general characters of phylum mollusca
5. What are mammals? Name its sub-classes. Give the characters and examples of each sub class.

EXERCISE

1. Encircle the correct choice

(i) From where Plants primarily obtain essential nutrients:
(a) Water (b) Air
(c) Soil (d) Light

(ii) Identify a micro nutrient for plants amongst the following:
(a) Potassium (b) Phosphorus
(c) Iron (d) Sluphur

(iii) In which mineral deficient soil carnivorous plant typically live:
(a) Nitrogen (b) Calcium
(c) Magnesium (d) Potassium

(iv) The plants growing in salt marshes close to sea are termed as:
(a) Hydrophytes (b) Xerophytes
(c) Mesophytes (d) Halophytes

(v) The cause of color of leaf turns in to dull or bluish green is the deficiency of:
(a) Nitrogen (b) Phosphate
(c) Potassium (d) Magnesium

(vi) Which principle type of auxin of higher plants synthesized at the apices of stem and root:
(a) PAA (b) 24-D
(c) IAA (d) NAA

(vii) Active transport of sucrose into sieve elements is a step commonly called:
(a) Phloem loading (b) unloading
(c) Diffusion (d) Osmosis

(viii) The capacity of a living system to lose water is:
(a) Osmotic pressure (b) Water potential
(c) Osmosis (d) Plasma

(ix) The production of lateral appendages depends on the activity of:
(a) Lateral meristem (b) lutercalary meristem
(c) Apical meristem (d) none of these

(x) Which process involved in the promotion of flowering by cold treatment:
(a) Photoperiodism (b) Vernalization
(c) Secondary Growth (d) Transpiration



EXERCISE

1. Encircle the correct choice.

(i) The process of nutrition does not include this:
(a) Taking in of food (b) Breaking down of food
(c) Absorption of nutrients (d) Removal of undigested food

(ii) The chemicals playing the most important role in digestion of food are:
(a) Hormones (b) Enzymes
(c) HCl (d) Both a & b
(e) None of these

(iii) The thecodont condition refers to:
(a) Presence of two sets of teeth during life time
(b) all alike teeth
(c) Sharp, pointed teeth
(d) Embedded teeth

(iv) The type of digestion in Planaria is:
(a) Intracellular (b) Extracellular
(c) Intercellular (d) Both a & b

(v) Number of milk teeth is:
(a) 10 (b) 15
(c) 20 (d) 25

(vi) Teeth meant for cutting are:
(a) Incisors (b) Canines
(c) Pre-molars (d) Molars

(vii) The part of gut also associated with respiratory system is:
(a) Stomach (b) Esophagus
(c) Pharynx (d) Both b & c

(viii) The number of salivary glands in our oral cavity are:
(a) 3 (b) 4
(c) 5 (d) 6

(ix) The rugae are related to:
(a) Stomach (b) Duodenum
(c) Jejunum (d) Ilium

(x) BMI over 30 kg/m^2 is considered as:
(a) Normal (b) Over-weight
(c) under-weight (d) Obesity

EXERCISE

1. Encircle the correct choice.

(i) Lymph most closely resembles which of the following?
(a) Blood (b) Urine
(c) Plasma (d) Interstitial fluid

(ii) What event initiates blood clotting?
(a) Contact with an irregular surface by platelets and other factors in plasma.
(b) Production of the enzyme thrombin
(c) Conversion of fibrinogen into fibrin
(d) Excess flow of blood through a capillary

(iii) Damage to the S.A Node in humans
(a) Is a major contributor to heart attacks.
(b) Would block conductance between the bundle branches and the purkinje fibers.
(c) Would have a negative effect on peripheral resistance
(d) Would disrupt the rate and timing of cardiac muscle contractions.

(iv) Which of the following is measured by an electrocardiogram?
(a) Impulses from the AV node
(b) Impulses of the parasympathetic nervous system that control heartbeat
(c) The speed on impulses from the SA node
(d) Contraction of the two atria

(v) Where are semilunar valves to be found in the mammalian heart?
(a) Where blood goes from atria to ventricles
(b) On the right side of the heart only
(c) Where the pulmonary veins attach to the heart
(d) At the places where blood leaves via the aorta and pulmonary arteries

(vi) Organisms in which a circulating body fluid is distinct from the fluid that directly surrounds the body's cells are likely to have which of the following?
(a) An open circulatory system
(b) A closed circulatory system
(c) A gastrovascular cavity
(d) Branched tracheae

(vii) Average blood pressure is lowest in which structure (s)
(a) Aorta (b) Arterioles
(c) Capillaries (d) Vena cava

(viii) Which of the following is/ are a cause (s) of vasoconstriction?
(a) Lying down after standing
(b) Standing after lying down
(c) Stress or hormone concentration
(d) Histamine secretion

(ix) Blood returning to the human heart in a pulmonary vein drains first into the?
(a) Left atrium (b) Right atrium
(c) Left ventricle (d) Right ventricle

(x) Which of the following is NOT an important function of the human circulation?
(a) Transport of nutrients and respiratory gases
(b) Regulation of body temperature
(c) Protection of the body by circulating antibodies
(d) Removal of waste products for excretion from the body

2. Write short answers of the following questions:

1. Why higher animals need circulatory system?
2. Why SA node is called pace maker of heart?
3. Why the capillaries have a single layer of endothelium?
4. What are LUB and DUB?
5. Why circulatory system of arthropods is called open type?
6. Why blood flow faster in arteries?
7. Distinguish between single circuit circulation and double circuit circulation.

3. Write detailed answers of the following questions:

1. Describe the structure of blood vessels in man.
2. Describe the structure of human heart.
3. What is circulation? Explain needs of circulation
4. State the phases of heartbeat in man.
5. Describe the flow of blood through human heart as regulated by the valves.
6. Describe the lymphatic system of man
7. Explain the electrocardiogram with the help of diagram.
8. Describe pulmonary circulation and systemic circulation



SUMMARY

- Immune system is the defence system of an organs consists of cells and proteins.
- Study of functioning and disorders of immune system is termed as immunology.
- The non-specific type of immune system against any pathogen which tries to enter in the body called innate-immune system.
- Adaption immune system is the specific types of immune system develop against specific foreign body called antigen.
- Innate immune system is consist of 1st line of defence and 2nd line of defence
- 1st line of defence is consist of skin, mucous membrane, ciliated epithelium, saliva, tears, mucous, gastric juice etc.
- 2nd line of defence consist of phagocytes (neutrophils monocytes), dendritic cells, cytoxic T cell, natural kill cells (N.K.C) perforin protein and granzyme.
- 3rd line of defence consist of lymphocyte T and B antibodies proteins.
- Active immunity is the consequence of individuals own immune response.
- Passive immunity is the kind of immunity. Where antibodies on immune cells produced by one individual are transferred to another individual to develop immunity.
- Allergies, auto immune diseases, transplant rejections are the disorder of immune system.

EXERCISE

1. Encircle the correct choice

(i) The study of functioning and disorders of immune system is termed as:
(a) Immunity (b) Immune response
(c) Immunology (d) all of these

(ii) It is responsible for specific immune response:
(a) Innate Immune system (b) First line of defense
(c) Second line of defense (d) Third line of defense

(iii) Following serves as physical barrier:
(a) Macrophage (b) Mucous membrane
(c) HCl (d) Saliva

(iv) Our digestive tract consists of what kind of barriers?
(a) Physical barriers (b) Biochemical barriers
(c) Both a & b (d) None of these

(v) Which of the following is involved in phagocytosis of bacteria?
(a) Neutrophil (b) Macrophage
(c) Both a & b (d) None of these

(vi) It is not involved in antigen presentation:
(a) Macrophage (b) Dendritic cell
(c) Neutrophil (d) Both a & b

(vii) The type of receptor exhibited by all nucleated cells other than macrophages is:
(a) MHC-I (b) MHC-II (c) Both a & b (d) TCR

(viii) Any foreign substance that can elicit immune response is called:
(a) Antibody (b) Antigen
(c) Both a & c (d) Antihistamine

(ix) The type of immunity through the transfer of antibodies from mother to her fetus is:
(a) Natural Active Immunity (b) Natural Passive Immunity
(c) Artificial Active Immunity (d) Artificial Passive Immunity

(x) Antibodies are secreted by:
(a) T cells (b) B cells
(c) Plasma cells (d) Macrophages

2. Write short answers of the following questions:

1. List out any six biochemical barriers.
2. How the tumor cells are dealt with by our immune system?
3. Differentiate between NK Cell and T_C cell.
4. List out the ways of second line of defense.
5. Even though the core proteins are the same, how antibodies differ from each other?
6. What is antipyretic therapy and why it is used?
7. List out four autoimmune disorders of man.
8. What is phagocytosis? Name some WBCs acting as phagocytes.
9. What is inflammation?
10. Outline the harmful effects of fever.

3. Write detailed answers of the following questions:

1. State and explain how a bacterium is identified as non-self by our immune system.
2. What is cytotoxicity? Compare and contrast the mechanisms of NK cell and T_C cell.
3. What is inflammation? Why it is developed and how it is controlled?
4. Explain the underlying mechanism of the fever. State and explain its protective role as well as harmful effects.
5. Discuss the structural features of antibody? How it is helpful in getting rid of pathogenic organisms.
6. What do you understand by autoimmune disorder? Explain the role of T cells and B cells in transplant rejection.



EXERCISE

1. Encircle the correct choice

(i) In which part of the respiratory system, gaseous exchange takes place?

(a) Alveoli (b) Pharynx
(c) Larynx (d) Trachea

(ii) Which of the following statements is true about involuntary breathing?

(a) It is controlled by the bronchioles
(b) It is controlled by the pulmonary arterioles
(c) It is controlled by the alveolar-capillary network
(d) It is controlled by the neurons, located in the medulla and pons

(iii) The tiny air sacs present in human lungs is called.

(a) Alveoli (b) Bronchus
(c) Bronchioles (d) All of the above

(iv) The exchange of gases between the external environment and the lungs.

(a) Respiration (b) External respiration
(c) Cellular respiration (d) None of the above

(v) The maximum volume of air contained in the lung by a full forced inhalation is called.

(a) Tidal volume (b) Vital capacity
(c) Ventilation rate (d) Total lung capacity

(vi) Which one of the following is correct regarding larynx?

(a) It houses the vocal cords
(b) It prevents the invading pathogens into the trachea
(c) It is an organ made of cartilage and connects the pharynx to the trachea
(d) All of the above.

(vii) Which of the following is the function of the trachea?

(a) Gaseous Exchange
(b) Filters the air we breathe
(c) Exhales the air from the body
(d) All of the above

(viii) Which of the following organs functions as an air conditioner?

(a) Larynx (b) Pharynx
(c) Nasal chambers (d) All of the above

(ix) Which of the following statements is true about the entry of air into the lungs?

(a) Air enters the body and travels to the lungs through the mouth and the nose

(b) Air enters the body and travels to the lungs through the esophagus and gullet

(c) Air enters the body and travels to the lungs through the windpipe and the pores

(d) Air enters the body and travels to the lungs through the nose and the nervous system.

2. Write short answers of the following questions:

1. When breathing process develop in animals?
2. What type of ventilation occur in human?
3. Why rate of breathing increases in human beings?
4. Why breathing of human being called negative pressure breathing?
5. How CO_2 from cell transport to lungs?
6. Which of the pigment is called respiratory pigment?
7. Draw a flow chart for the passage of air from external nares to alveoli.
8. Why hair and mucus glands are present in nostrils and trachea?
9. How exchange of gases occurs at the alveolar level?

3. Write detailed answers of the following questions:

1. What is respiration? Describe human respiratory system?
2. What are the 5 main functions of the respiratory system?
3. Draw labelled diagram of human respiratory system.
4. Describe the three regions of the pharynx and their functions.
5. What is the mechanism of inspiration and expiration?
6. Describe three ways in which carbon dioxide can be transported.